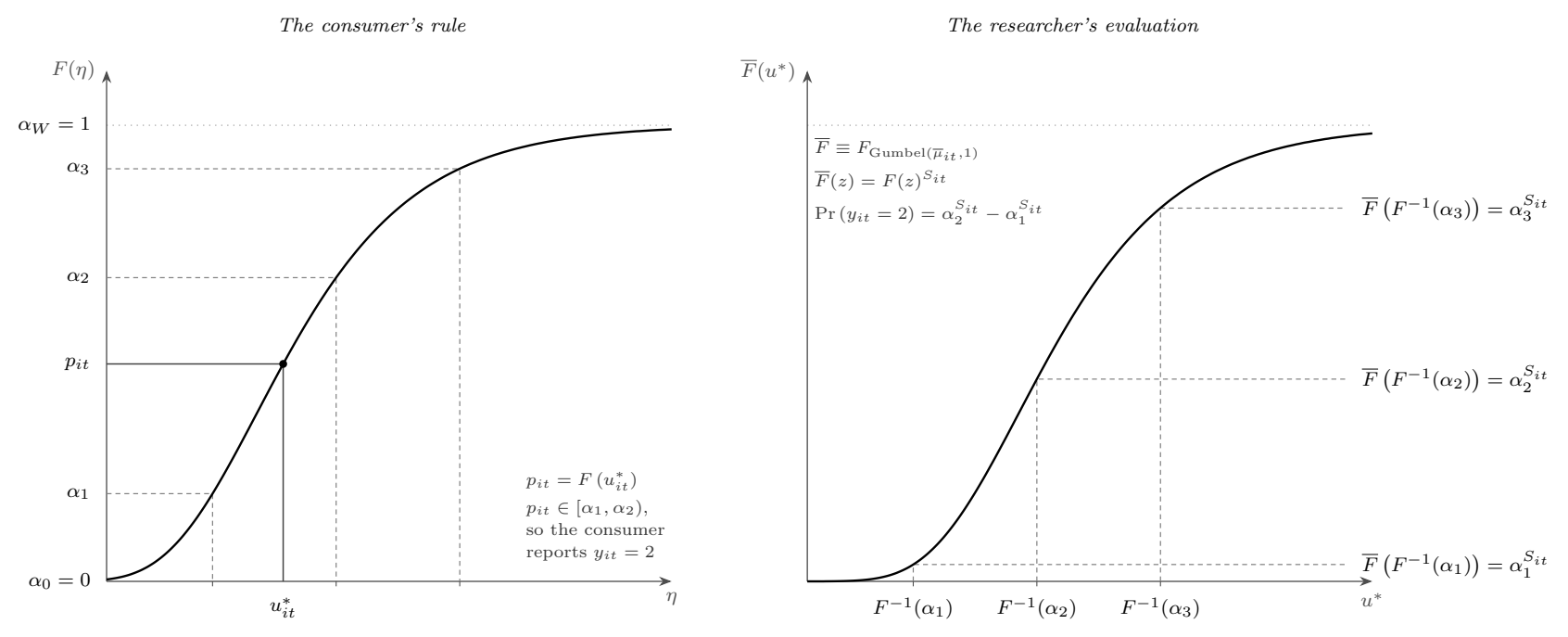

The consumer's rule
$F(\eta)$
$\alpha_W = 1$
α_3
α_2
p_{it}
α_1
$\alpha_0 = 0$
u^*_{it}
η
$p_{it} = F(u^*_{it})$
$p_{it} \in [\alpha_1, \alpha_2)$,
so the consumer
reports $y_{it} = 2$
The researcher's evaluation
$\overline{F}(u^*)$
$\overline{F} \equiv F_{\text{Gumbel}(\overline{\mu}_{it},1)}$
$\overline{F}(z) = F(z)^{S_{it}}$
$\Pr(y_{it} = 2) = \alpha_2^{S_{it}} - \alpha_1^{S_{it}}$
$\overline{F}\left(F^{-1}(\alpha_3)\right) = \alpha_3^{S_{it}}$
$\overline{F}\left(F^{-1}(\alpha_2)\right) = \alpha_2^{S_{it}}$
$\overline{F}\left(F^{-1}(\alpha_1)\right) = \alpha_1^{S_{it}}$
$F^{-1}(\alpha_1)$
$F^{-1}(\alpha_2)$
$F^{-1}(\alpha_3)$
u^*